

Probability & Sample Space – Answer Sheet

WorksheetHub AI
Mathematics · Grade
7

Unit 4 · Data & Probability

Medium

Probability

A. Single events

- 1 Primes are 2, 3, 5 $\rightarrow 3/6 = 1/2$
- 2 5 blue out of 10 $\rightarrow 5/10 = 1/2$
- 3 $P(\text{yellow}) = 2/8 = 1/4$; $P(\text{red or blue}) = 4/8 = 1/2$

B. Two dice (use the sample space table)

- 4 6 outcomes $\rightarrow 6/36 = 1/6$
- 5 10:3, 11:2, 12:1 $\rightarrow 6$ outcomes $\rightarrow 6/36 = 1/6$
- 6 0 – it is impossible (the smallest total is 2)

C. Experimental probability

- 7 $8 \div 50 = 0.16$
- 8 $1/6 \times 120 = 20$
- 9 Not definitely – 20 heads is expected, 26 is a bit high but could happen by chance. Flip more times to be surer.

D. Listing outcomes

- 10 H1 H2 H3 H4 H5 H6 T1 T2 T3 T4 T5 T6 $\rightarrow 12$ outcomes
- 11 H2, H4, H6 $\rightarrow 3/12 = 1/4$

★ Challenge

- 12 $P(\text{blue}) = 0.6$, so 12 is 60% $\rightarrow \text{total} = 20$
- 13 7 is the most likely: $6/36 = 1/6$